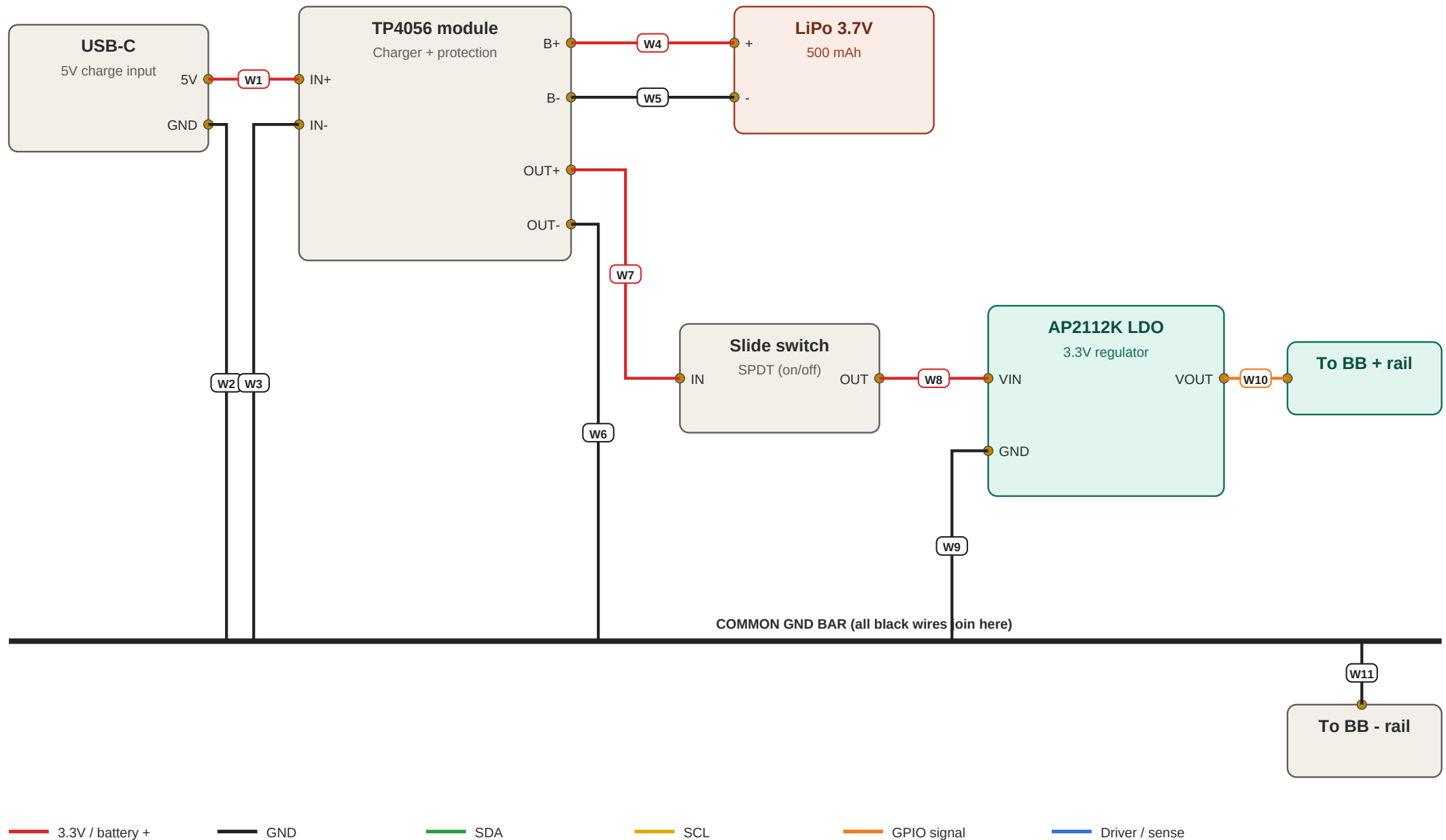


Power section - module wiring

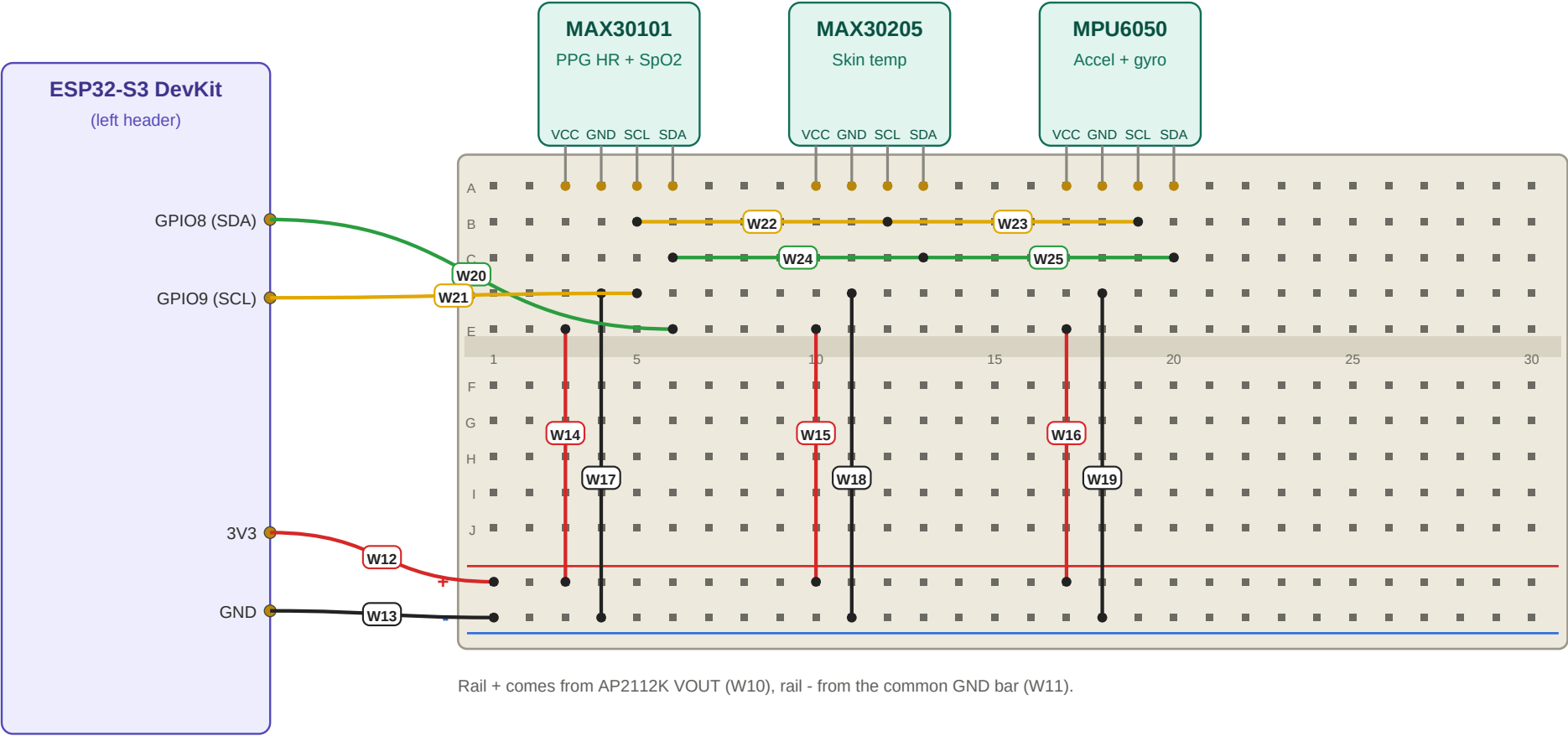
Page 1 of 6 | Charger, battery, switch and 3.3V regulator (wire numbers W1-W11 match the wire table)



Note: TP4056 B- and OUT- are the same net inside the module. Tap the battery positive (B+) for the divider on Breadboard 2 (wire W39).

Breadboard 1 - sensors and I2C bus

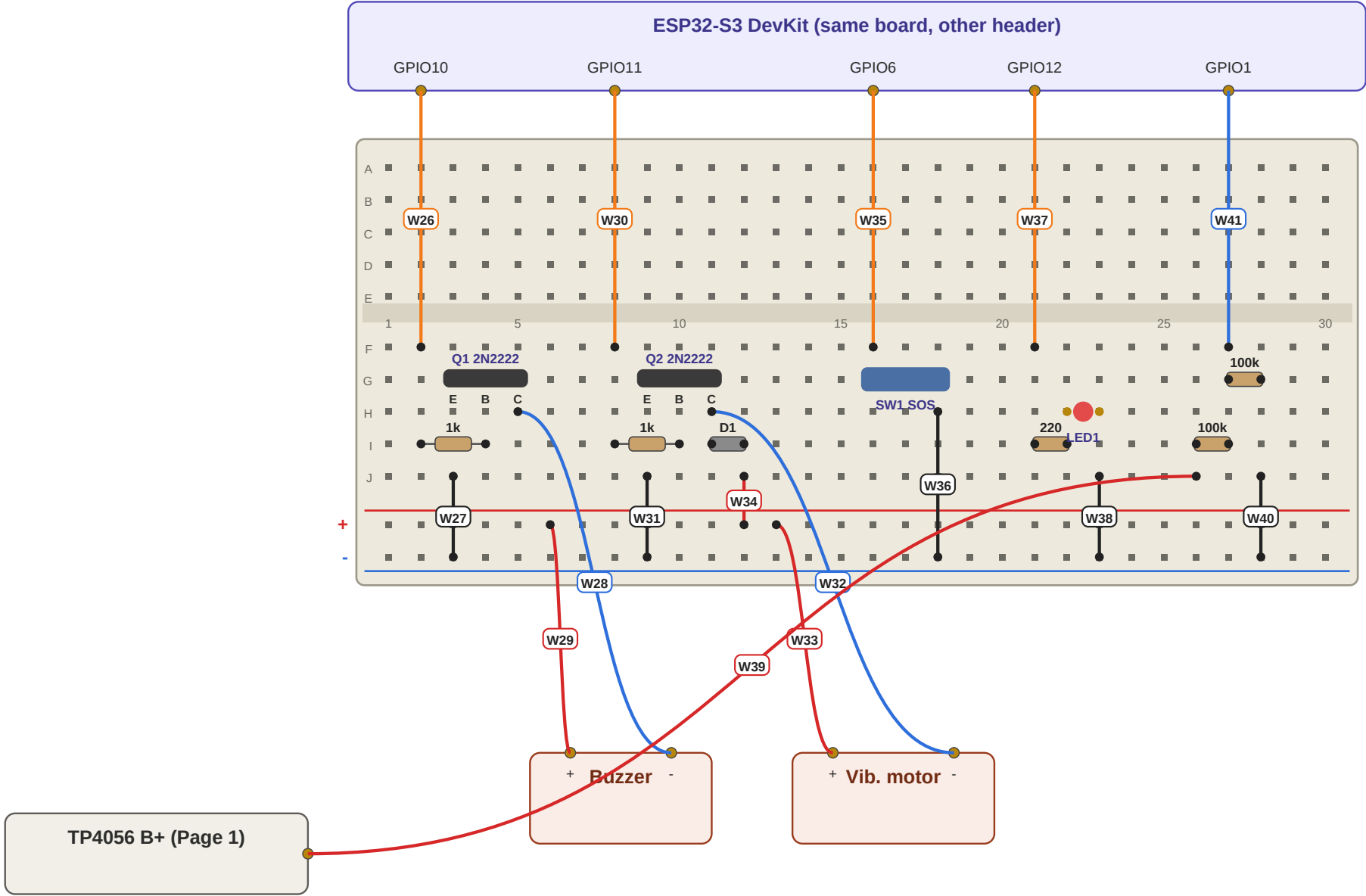
Page 2 of 6 | ESP32-S3 with MAX30101, MAX30205 and MPU6050 on one shared I2C bus



3.3V / battery + GND SDA SCL GPIO signal Driver / sense

Breadboard 2 - buttons, alerts and battery monitor

Page 3 of 6 | SOS button, buzzer and motor drivers, status LED, battery divider



3.3V / battery + GND SDA SCL GPIO signal Driver / sense

Wire table - which two points each wire connects

Page 4 of 6 | W1-W20 (power, breadboard 1 rails, first I2C wires)

Wire	Colour	From (point 1)	To (point 2)	Purpose
W1	Red	USB-C 5V	TP4056 IN+	5V charging input
W2	Black	USB-C GND	Common GND bar	Ground return
W3	Black	TP4056 IN-	Common GND bar	Ground return
W4	Red	TP4056 B+	LiPo battery +	Charge / discharge path
W5	Black	TP4056 B-	LiPo battery -	Charge / discharge path
W6	Black	TP4056 OUT-	Common GND bar	Load ground
W7	Red	TP4056 OUT+	Slide switch IN	Protected battery output
W8	Red	Slide switch OUT	AP2112K VIN	Switched supply
W9	Black	AP2112K GND	Common GND bar	Regulator ground
W10	Orange	AP2112K VOUT	Breadboard + rail (3.3V)	3.3V supply
W11	Black	Common GND bar	Breadboard - rail	Ground rail
W12	Red	Breadboard + rail (col 1)	ESP32-S3 3V3 pin	Power the MCU
W13	Black	ESP32-S3 GND pin	Breadboard - rail (col 1)	MCU ground
W14	Red	Rail + (col 3)	MAX30101 VCC (E3)	Sensor power
W15	Red	Rail + (col 10)	MAX30205 VCC (E10)	Sensor power
W16	Red	Rail + (col 17)	MPU6050 VCC (E17)	Sensor power
W17	Black	Rail - (col 4)	MAX30101 GND (D4)	Sensor ground
W18	Black	Rail - (col 11)	MAX30205 GND (D11)	Sensor ground
W19	Black	Rail - (col 18)	MPU6050 GND (D18)	Sensor ground
W20	Green	ESP32-S3 GPIO8	MAX30101 SDA (E6)	I2C data

Wire table - which two points each wire connects

Page 5 of 6 | W21-W41 (I2C bus jumpers, buttons, drivers, LED, battery monitor)

Wire	Colour	From (point 1)	To (point 2)	Purpose
W21	Yellow	ESP32-S3 GPIO9	MAX30101 SCL (D5)	I2C clock
W22	Yellow	MAX30101 SCL (B5)	MAX30205 SCL (B12)	I2C clock bus
W23	Yellow	MAX30205 SCL (B12)	MPU6050 SCL (B19)	I2C clock bus
W24	Green	MAX30101 SDA (C6)	MAX30205 SDA (C13)	I2C data bus
W25	Green	MAX30205 SDA (C13)	MPU6050 SDA (C20)	I2C data bus
W26	Orange	ESP32-S3 GPIO10	R1 1k (hole F2)	Buzzer driver input
W27	Black	Q1 emitter (J3)	Rail - (col 3)	Driver ground
W28	Blue	Q1 collector (H5)	Buzzer -	Switch buzzer to GND
W29	Red	Buzzer +	Rail + (col 6)	Buzzer supply
W30	Orange	ESP32-S3 GPIO11	R2 1k (hole F8)	Motor driver input
W31	Black	Q2 emitter (J9)	Rail - (col 9)	Driver ground
W32	Blue	Q2 collector (H11)	Vibration motor -	Switch motor to GND
W33	Red	Vibration motor +	Rail + (col 13)	Motor supply
W34	Red	D1 cathode (J12)	Rail + (col 12)	Flyback diode to 3.3V
W35	Orange	ESP32-S3 GPIO6	SOS button leg 1 (F16)	SOS input
W36	Black	SOS button leg 2 (H18)	Rail - (col 18)	Button to GND
W37	Orange	ESP32-S3 GPIO12	R3 220 ohm (hole F21)	LED drive
W38	Black	LED cathode (J23)	Rail - (col 23)	LED ground
W39	Red	TP4056 B+	R4 100k (hole J26)	Battery sense tap
W40	Black	R5 100k (hole J28)	Rail - (col 28)	Divider ground
W41	Blue	ESP32-S3 GPIO1 (ADC)	Divider midpoint (F27)	Battery voltage read

Component placement on the breadboards

Page 6 of 6 | Hole names use column number + row letter (example: E3 = column 3, row E)

Component	Pin / leg	Breadboard hole	Board
MAX30101	VCC, GND, SCL, SDA	A3, A4, A5, A6	Breadboard 1
MAX30205	VCC, GND, SCL, SDA	A10, A11, A12, A13	Breadboard 1
MPU6050	VCC, GND, SCL, SDA	A17, A18, A19, A20	Breadboard 1
Q1 2N2222 (buzzer)	E, B, C	G3, G4, G5	Breadboard 2
R1 1k (Q1 base)	Leg 1, leg 2	I2, I4	Breadboard 2
Q2 2N2222 (motor)	E, B, C	G9, G10, G11	Breadboard 2
R2 1k (Q2 base)	Leg 1, leg 2	I8, I10	Breadboard 2
D1 1N4148	Anode, cathode (band)	I11, I12	Breadboard 2
SW1 SOS button	Leg 1, leg 2	G16, G18	Breadboard 2
R3 220 ohm	Leg 1, leg 2	I21, I22	Breadboard 2
LED1 status	Anode, cathode	H22, H23	Breadboard 2
R4 100k (divider top)	Leg 1, leg 2	I26, I27	Breadboard 2
R5 100k (divider bottom)	Leg 1, leg 2	G27, G28	Breadboard 2

a pair from SDA and SCL to 3.3V only if the bus is unreliable. Keep the MAX30101 and MAX30205 in contact with the skin on the underside of the arm band.
Notes: 2N2222 pinout shown is E-B-C (flat side facing you, PN2222 type). Check your transistor datasheet. Most sensor breakouts already have 4.7k I2C pull-ups; add